

GREGORY S. FITZGERALD, PhD

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PROFILE

Behavioral neuroscientist and quantitative researcher with preclinical and analytical experience across academic and industry-adjacent research. First-author and contributing-author publications spanning Alzheimer's therapeutics, whole-brain imaging, and applied machine learning. Methodological focus on systematic evidence synthesis, research-integrity tooling, and reproducible statistical pipelines.

EDUCATION

Ph.D., Behavioral Neuroscience | State University of New York at Albany
2026

B.A., Psychology and Neuroscience, *Magna Cum Laude* | Queens College, City University of New York (minor: Japanese)
2013

RESEARCH & PROFESSIONAL EXPERIENCE

Doctoral Research Scientist | State University of New York at Albany, Stellar Lab
2016 - 2026

- Ran an independent preclinical research program testing IGF-2 (a neurotrophic factor) and metabolic interventions in rodent Alzheimer's models; first-author publication in *CNS Neuroscience & Therapeutics* (2023).
- Built an automated evidence-synthesis pipeline in R and Python for a systematic cross-species meta-analysis; identified a 2.5x effect-size gap between rodent and human cognitive findings.
- Executed the full experimental pipeline: intranasal compound administration, cognitive phenotyping (fear, spatial, and working memory), hippocampal cell culture, ICC, ELISA, and Western blot.

Laboratory Technician & Manager | Cold Spring Harbor Laboratory, Osten Lab
2011 - 2016

- Contributing author to a *Cell* publication (2017): validated transgenic mouse lines with IHC and contributed to serial two-photon tomography for whole-brain, single-cell-resolution mapping of GABAergic neuron distribution.
- Studied the neurobiology of social behavior in *Cntnap2* knockout autism models via stereotaxic viral injections, confocal microscopy, and IHC.
- Promoted from technician to lab manager: ran the animal colony, genotyping pipeline, procurement, and budgets, and coordinated across postdoctoral research teams.

Adjunct Instructor, Statistics | SUNY Albany & Hudson Valley Community College
2019 - 2026

- Primary instructor for undergraduate statistics across six years with excellent student evaluations; designed curriculum, assessments, and grading; mentored 16 undergraduate researchers.
- Faculty panel presenter, HVCC Faculty Workshop Day (March 5, 2025): led the "Interactive Documents How-To" segment of the session "What Can I Do With Feedback Fruits?", demonstrating for colleagues the interactive-document tool HVCC was piloting.

Undergraduate Research Assistant | Bodnar Lab, Queens College, City University of New York
2009 - 2011

- Co-authored two peer-reviewed publications on reward-circuitry neuropharmacology; mapped brain activation via c-Fos immunohistochemistry in mesocorticolimbic regions.

Founder | EE Performance
2021

- Preclinical-stage biotech concept built around exogenous exerkines as exercise mimetics for Alzheimer's disease; developed the scientific and commercial rationale, competed in the UAlbany Blackstone LaunchPad venture competition, and completed a patent-landscape analysis.

Research Software Developer | Metascience Observatory (Mind First Foundation)
2025 - present

- Lead developer of the automated meta-analysis pipeline behind a cross-field assessment of research replicability; define its effect-size extraction and data-quality standards. Co-author on the project's methods paper.
- Co-developer, Forensic Metascience Toolkit: a system that checks whether the statistics a paper reports are internally possible. Built the classification layer determining which statistical checks validly apply to which reported values, and closed several pathways that could have produced false accusations against sound papers.
- Replication database: cleaned and reconciled datasets ingested from multiple external sources so that studies were commensurate, and built the conversion tools placing reported p-values onto a common effect-size scale with confidence intervals -- the standardization that made cross-field synthesis possible.
- Built Figure Extractor, a machine-learning system that classifies charts in published papers and extracts their underlying numerical data; and Meta-Reference Toolkit, a Python package that cleans and de-duplicates large reference libraries.

PEER-REVIEWED PUBLICATIONS

1. Kim Y, Yang GR, Pradhan K, Venkataraju KU, Bota M, Garcia del Molino LC, **Fitzgerald G**, Ram K, He M, Levine JM, Mitra P, Huang JZ, Wang X-J, Osten P. Brain-wide Maps Reveal Stereotyped Cell-Type-Based Cortical Architecture and Subcortical Sexual Dimorphism. *Cell*. 2017;171(2):456-469.
2. **Fitzgerald GS**, Chuchta TG, McNay EC. Insulin-like growth factor-2 is a promising candidate for the treatment and prevention of Alzheimer's disease. *CNS Neuroscience & Therapeutics*. 2023.
3. Ralbovsky NM, **Fitzgerald GS**, McNay EC, Lednev IK. Raman spectroscopy and machine learning for Alzheimer's disease screening. *Spectrochimica Acta Part A*. 2021.
4. Kraft TT, Yakubov Y, Huang D, **Fitzgerald G**, Natanova E, Sclafani A, Bodnar RJ. Dopamine D1 and opioid receptor antagonists differentially reduce the acquisition and expression of fructose-conditioned flavor preferences in BALB/c and SWR mice. *Physiology & Behavior*. 2015;151:213-220.
5. Kraft TT, Yakubov Y, Huang D, **Fitzgerald G**, Acosta V, Natanova E, Touzani K, Sclafani A, Bodnar RJ. Dopamine D1 and opioid receptor antagonism effects on the acquisition and expression of fat-conditioned flavor preferences in BALB/c and SWR inbred mice. *Pharmacology Biochemistry and Behavior*. 2013;110:127-136.

PREPRINTS & WORKS IN PROGRESS

- Fitzgerald GS. A Cross-Species Meta-Analysis of Environmental Enrichment Effects on Cognition. Preprint, 2026.

PUBLIC & POPULAR WRITING

- Fitzgerald G, Elton D. "WTH is Cerebrolysin, actually?" *More is Different* (Substack), August 5, 2024.

SELECTED CONFERENCE PRESENTATIONS

- Fitzgerald GS, Sullivan EG, Ascoti M, McNay EC. Does phosphorylation of the insulin-like growth factor receptor (IGF2R) correlate with IGF2 binding activity in rat brain tissue? CURCE 19th Annual Student Symposium, University at Albany, 2022.
- Fitzgerald GS, Fumo EA, Hatase K, Sullivan E, Siskind A, Young H, Matthews A, McNay EC. Medial frontal cortex lidocaine administration and set-shift performance. NEURON Conference, 2020.
- Fitzgerald GS, Mallon LP. Molecular effectors of insulin's role in hippocampal memory processing across stages of memory. Society for Neuroscience Annual Meeting, 2018 (Program 084.02).
- Fitzgerald G, Kim Y, Osten P. Cntnap2 knockout mice exhibit reduced sociability in the three-chamber test. NEURON / Hunter Psychology Conference; URME Research Symposium; Sigma Xi Research Symposium, 2012.
- Kim Y, Venkataraju KU, Pradhan K, Fitzgerald G, He M, Levine J, Huang Z, Osten P. Quantitative brain-wide mapping of GABAergic neuronal subtypes in mice. Society for Neuroscience Annual Meeting, 2015 (571.02).
- Kim Y, Pradhan K, Fitzgerald G, Venkataraju KU, Osten P. Serial two-photon tomography-based whole-brain activity mapping in the Cntnap2 KO mouse model of autism. Society for Neuroscience Annual Meeting, 2013 (719.04).
- Kraft TT, Yakubov Y, Huang D, Fitzgerald G, Acosta V, Natanova E, Touzani K, Sclafani A, Bodnar RJ. Differential dopamine D1 and opioid receptor modulation of fat-conditioned flavor preferences in BALB/c and SWR inbred mice. Society for Neuroscience Annual Meeting, 2012.

- Dela Cruz J, Coke T, Sampson C, Fitzgerald G, Kest K, Icaza-Cukali D, Rotella FM, Ranaldi R, Touzani K, Sclafani A, Bodnar RJ. c-Fos induction in the rat amygdala following oral intake and intragastric infusion of sugar. Society for Neuroscience Annual Meeting, 2011; IBRO World Congress (Florence, Italy), 2011; Einstein in the City, 2011.
- Kraft TT, Fitzgerald G, Yakubov Y, Avizova O, Huang D, Faulding BD, Stern R, Touzani K, Sclafani A, Bodnar RJ. Flavor preferences conditioned by fat in BALB/c and SWR inbred mice. Society for Neuroscience Annual Meeting, 2011.
- Yakubov Y, Fitzgerald G, Acosta V, Borukhova N, Borukhov L, Elkomos-Botros M, Luu L, Kraft TT, Bodnar RJ. Dopamine and opioids modulate fructose-conditioned flavor preferences in BALB/c and SWR inbred mice. NEURON Conference, 2011; Society for Neuroscience Annual Meeting, 2010.

HONORS & AWARDS

- Magna Cum Laude, Queens College, CUNY (2013)
- Undergraduate Research Program (URP) Fund Scholar, Cold Spring Harbor Laboratory (2011)
- Undergraduate Research & Mentoring Education (URME) Scholar, Queens College, CUNY (2010)
- Salutatorian, Francis Lewis High School (2009)

TECHNICAL SKILLS

Computational & Quantitative: R (tidyverse, metafor, Quarto), Python (pandas, NumPy), SQL & SQLite, Advanced Excel, SPSS, GraphPad Prism | **Meta-Analysis & Systematic Review** | **Mixed-Effects & Bayesian Modeling** | **Applied Machine Learning** | **Reproducible Pipelines & Data Visualization**

Laboratory: Stereotaxic Neurosurgery & Viral Vector Delivery | Microdialysis | Rodent Behavioral Testing | IHC/ICC | Confocal & Two-Photon Microscopy | Western Blot | ELISA | Primary Cell Culture | Colony Management & IACUC